1. Title

SecureVault : A Full-Stack Password Vault & Credential Management System

2. Objective

Build a full-stack web application (React.js frontend + Spring Boot backend) that enables users to securely store, organize, manage, generate, share, and monitor credentials through encrypted vaults. The platform provides password management, multi-factor authentication, secure credential sharing, security monitoring, threat detection, audit logging, and password health analytics through a centralized dashboard. This solution can be used for personal password management, enterprise credential storage, team collaboration, and cybersecurity applications.

Outcomes

(i) Developed and deployed a secure password vault and credential management platform.

(ii) Implemented secure authentication and multi-factor authentication workflows.

(iii) Built encrypted password storage and credential management systems.

(iv) Developed password generation and password strength analysis modules.

(v) Implemented secure credential sharing and permission management workflows.

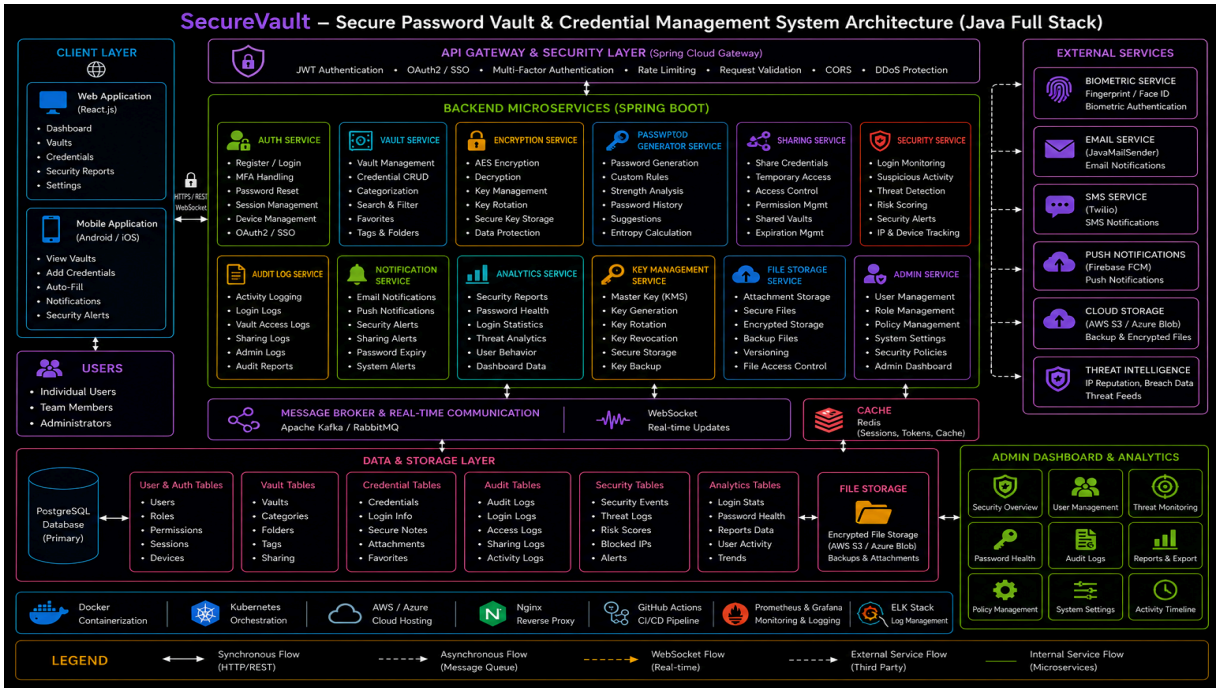
(vi) Built security monitoring and suspicious activity detection systems.

(vii) Developed analytics dashboards for password health and security monitoring.

(viii) Deployed the platform using Docker and cloud deployment platforms.

(ix) Implemented audit logging and threat monitoring systems.

3. Architecture Components



4. Modules to be Implemented

1. User Authentication & Access Control Module

(i) User registration

(ii) User login

(iii) Multi-factor authentication

(iv) Password recovery

(v) Session management

(vi) OAuth2 login

(vii) Device management

Roles

- User
- Team Member
- Administrator

2. Password Vault Module

(i) Secure credential storage

(ii) Credential categorization

(iii) Vault organization

(iv) Credential search

(v) Credential filtering

(vi) Favorites management

(vii) Secure notes storage

Credential Types

- Website Login
- Email Accounts
- Banking Credentials
- Social Media Accounts
- Application Accounts
- API Keys
- Secure Notes

3. Password Generator Module

(i) Strong password generation

(ii) Custom password rules

(iii) Password length configuration

(iv) Special character support

(v) Password strength analysis

(vi) Password suggestions

4. Secure Sharing Module

(i) Credential sharing

(ii) Temporary access sharing

(iii) Team vault management

(iv) Access permissions

(v) Role-based sharing

(vi) Shared credential expiration

Permission Levels

- View Only
- Edit Access
- Full Management

5. Encryption & Security Module

(i) AES encryption

(ii) Password hashing

(iii) Secure key generation

(iv) Key rotation

(v) Secure token management

(vi) API security

(vii) Data encryption at rest

(viii) Data encryption in transit

6. Security Monitoring Module

(i) Login monitoring

(ii) Suspicious activity detection

(iii) Device tracking

(iv) Threat monitoring

(v) Security alerts

(vi) Risk analysis

(vii) Login anomaly detection

7. Audit Logging Module

(i) Login logs

(ii) Vault access logs

(iii) Sharing activity logs

(iv) Security logs

(v) System activity tracking

(vi) Audit reports

8. Notification Module

(i) Login alerts

(ii) Security notifications

(iii) Sharing notifications

(iv) Password expiration alerts

(v) Email notifications

(vi) Push notifications

(vii) Risk alerts

9. Analytics Dashboard Module

User Dashboard

(i) Password health score

(ii) Credential overview

(iii) Security alerts

(iv) Recent activity

(v) Password strength reports

Team Dashboard

(i) Shared credentials

(ii) Team activity

(iii) Access monitoring

(iv) Security insights

(v) Audit reports

Admin Dashboard

(i) User management

(ii) Security monitoring

(iii) Threat analytics

(iv) Audit log management

(v) System monitoring

(vi) Compliance reports

10. Reports & Export Module

(i) Security reports

(ii) Password health reports

(iii) Audit reports

(iv) User activity reports

(v) Threat monitoring reports

(vi) PDF export

(vii) Excel export

11. Final Integration, Testing & Deployment

5. Week-wise Module Implementation and High-Level Requirements

Milestone 1: Week 1 & 2 — Project Initialization, Design Process & Core Setup

Tasks

(i) Define project objectives and security workflows.

(ii) Design secure system architecture and database schema.

(iii) Create UI wireframes and credential management workflow planning.

(iv) Setup React frontend and Spring Boot backend.

(v) Implement secure authentication and session management.

(vi) Build password vault management module.

(vii) Develop secure credential storage workflows.

Outcomes

(i) Understand secure authentication and password management workflows.

(ii) Learn secure system architecture and database design concepts.

(iii) Build frontend and backend project initialization.

(iv) Working authentication and password vault management system.

Milestone 2: Week 3 & 4 — Encryption System & Password Management

Tasks

(i) Implement AES encryption workflows.

(ii) Build password generation module.

(iii) Develop password strength checker.

(iv) Implement secure credential sharing workflows.

(v) Build permission and access control system.

Outcomes

(i) Implement secure encryption and credential protection systems.

(ii) Build password generation and sharing workflows.

(iii) Understand cybersecurity and access management concepts.

(iv) Generate secure password and credential management features.

Milestone 3: Week 5 & 6 — Security Monitoring & Analytics

Tasks

(i) Develop login monitoring system.

(ii) Build suspicious activity detection workflows.

(iii) Generate security alerts and audit logs.

(iv) Build security analytics dashboards.

(v) Generate password health and login activity reports.

Outcomes

(i) Build security monitoring and threat detection systems.

(ii) Implement analytics and reporting workflows.

(iii) Understand cybersecurity monitoring and alert systems.

(iv) Complete end-to-end credential security workflows.

Milestone 4: Week 7 & 8 — Testing, Deployment & Documentation

Tasks

(i) Perform security testing and workflow validation.

(ii) Improve UI responsiveness and security optimization.

(iii) Deploy platform using Docker and cloud environments.

(iv) Prepare final project documentation and presentation.

(v) Demonstrate the complete SecureVault platform.

Outcomes

(i) Gain deployment and security testing experience.

(ii) Improve platform stability and usability.

(iii) Complete live deployment and final demonstration.

(iv) Prepare professional project documentation and presentation.

6. Evaluation Criteria

Milestone 1 (Week 2)

(i) Project initialization and architecture setup completed.

(ii) Authentication and vault management implemented.

(iii) Credential storage workflows functional.

(iv) System design and UI planning completed.

Milestone 2 (Week 4)

(i) Encryption and password generation system implemented.

(ii) Secure sharing and access control workflows functional.

(iii) Password strength analysis working.

(iv) Credential protection workflows integrated.

Milestone 3 (Week 6)

(i) Security monitoring and alert system implemented.

(ii) Analytics dashboard and audit logging functional.

(iii) Threat detection workflows working.

(iv) Password health reports generated.

Milestone 4 (Week 8)

(i) Fully deployed frontend and backend.

(ii) Security testing and validation completed.

(iii) Documentation and presentation prepared.

(iv) Successful end-to-end platform demonstration completed.

7. Tools & Tech Stack

Programming Language

Backend

(i) Java

(ii) Spring Boot

Frontend

(i) JavaScript

(ii) React.js

Database

Primary Database

(i) PostgreSQL

Cache

(i) Redis

Authentication & Security

(i) Spring Security

(ii) JWT Authentication

(iii) OAuth2 Login

(iv) Multi-Factor Authentication (MFA)

(v) AES-256 Encryption

(vi) BCrypt Password Hashing

Libraries & Frameworks

Backend

(i) Spring Boot

(ii) Spring Security

(iii) Spring Data JPA

(iv) Hibernate

(v) Maven

(vi) Spring WebSocket

Frontend

(i) React.js

(ii) React Router

(iii) Axios

(iv) Tailwind CSS

(v) Redux Toolkit

Storage

(i) AWS S3

(ii) Encrypted File Storage

Notifications

(i) Firebase Cloud Messaging

(ii) JavaMailSender

Testing

(i) JUnit

(ii) Mockito

(iii) Postman

(iv) React Testing Library

Dev & Deployment Tools

(i) IntelliJ IDEA

(ii) VS Code

(iii) Git & GitHub

(iv) Docker

(v) Docker Compose

(vi) GitHub Actions

(vii) AWS / Azure

(viii) Nginx

(ix) Postman

8. Performance Metrics

Security Performance

(i) Authentication success rate

(ii) Encryption and decryption speed

(iii) Threat detection accuracy

(iv) MFA success rate

Vault Management Performance

(i) Credential retrieval speed

(ii) Password generation efficiency

(iii) Secure sharing reliability

(iv) Search performance

System Performance

(i) API response time

(ii) Concurrent user handling

(iii) Database query optimization

(iv) Cache performance

9. Example Quantitative Goals

Authentication Security

Achieve secure and reliable multi-factor authentication workflows.

Credential Protection

Maintain encrypted and secure password storage with fast retrieval performance.

Security Monitoring

Detect suspicious login and threat activities with high accuracy.

Platform Performance

Support multiple concurrent users with stable and secure application performance.